AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

EDUPULSE: AI-DRIVEN REAL-TIME STUDENT FEEDBACK ANALYTICS FOR SMART LEARNING

Project Domain

Excel + Power BI + Copilot /AI Chat bot

A Project Report

Submitted in partial fulfilment of the requirements of the AICW Project

by:

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ABSTRACT OF THE PROJECT

This project is part of the AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship is designed to equip students with practical, hands-on skills in Artificial Intelligence, Data Analytics, and Business Intelligence tools.

This project focuses on the application of data analytics and artificial intelligence tools to evaluate student feedback for the Cost Accounting course across B.Com specializations. The Pulse Point – Faculty Performance Ranking Board is a comprehensive analytical framework designed to evaluate and enhance teaching excellence within higher education institutions. The system integrates multiple performance dimensions—RI-Excellence Score, Improvement Area Score, and Rating Score—to provide a balanced and data-driven view of faculty effectiveness. The objectives of the study were to structure and clean the dataset, analyze faculty performance and course effectiveness using quantitative techniques, develop interactive dashboards using Microsoft Excel and Power BI, and integrate AI tools such as Microsoft Copilot and Chat bot systems for automated insight generation.

Through structured data collection from faculty performance metrics and student feedback, the system computes weighted scores that reflect both teaching quality and continuous improvement. These scores are processed to generate a Ranking Index, enabling transparent comparison among faculty members.

The resulting Faculty Performance Ranking Board serves as a decision-support tool for academic administrators, promoting accountability, recognition, and targeted mentoring. Ultimately, this project fosters a culture of continuous professional growth, aligning teaching excellence with institutional objectives and student satisfaction. It demonstrates how performance analytics can drive quality assurance and empower educators to achieve sustained academic impact.

The project concludes that the integration of business intelligence tools and AI-driven analytics can significantly improve the evaluation of academic performance and support data-driven decision-making in educational institutions. The approach demonstrates a scalable and efficient framework for enhancing teaching quality and student learning outcomes.

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CHAPTER 1
INTRODUCTION

The rapid advancement of data analytics and artificial intelligence has significantly transformed decision-making processes across various sectors, including education. In higher education institutions, student feedback serves as a critical tool for evaluating teaching effectiveness, course design, and overall learning outcomes. However, traditional methods of analyzing feedback are often manual, time-consuming, and lack the ability to generate actionable insights.

This project, titled **"Student Feedback Analytics for Smart Learning in Cost Accounting (B.Com)"**, focuses on leveraging data analytics and AI-enabled tools to systematically evaluate student feedback. The study is based on a structured dataset collected from 480 students across different B.Com specializations, including General, Accounting & Finance, Computer Applications, and Corporate Secretaryship. The feedback is categorized under key themes such as syllabus coverage, communication and methodology, assessment and feedback, growth opportunities, teaching support, and employability skills.

The primary challenge addressed in this project is the presence of unstructured and inconsistent data, which limits the effectiveness of traditional evaluation methods. By applying data cleaning techniques and analytical tools such as Microsoft Excel and Power BI, the project transforms raw feedback into meaningful insights. Furthermore, the integration of AI tools, including Microsoft Copilot and chatbot systems, enhances the analytical process by automating data interpretation and generating intelligent summaries.

The project adopts a business analytics approach to evaluate faculty performance, identify strengths and weaknesses in course delivery, and highlight areas requiring improvement. Interactive dashboards are developed to visualize key performance indicators, enabling stakeholders to make informed decisions based on data-driven evidence.

Overall, this project demonstrates how modern data analytics and AI technologies can be effectively applied in the academic domain to improve teaching quality, enhance student learning experiences, and support continuous institutional development.

1.1 Problem Statement

Educational institutions routinely collect student feedback to evaluate teaching effectiveness and course quality. However, in the context of the Cost Accounting course offered across B.Com specializations, the feedback data is often underutilized due to several operational and analytical limitations.

The primary issue lies in the **lack of a structured and systematic approach** to process and analyze student feedback. The collected data is typically raw, containing inconsistencies such as missing values, duplicate entries, and invalid responses, which reduces its reliability for decision-making. Additionally, traditional methods of analysis rely heavily on manual interpretation, making the process time-consuming, error-prone, and inefficient.

Another critical challenge is the **absence of integrated analytical tools and visualization mechanisms**. Without the use of business intelligence tools, institutions are unable to effectively identify patterns, compare faculty performance, or evaluate key dimensions such as communication, engagement, and assessment practices. As a result, actionable insights that could improve teaching quality and student learning outcomes remain largely unexplored.

Furthermore, there **adoption** is **limited of AI-driven tools** such as Microsoft Copilot and chatbot systems in academic feedback analysis. This restricts the ability to automate insight generation, detect trends, and support data-driven decision-making.

Therefore, the core problem addressed in this project is the need to **transform raw and unstructured student feedback data into meaningful, accurate, and actionable insights** using data cleaning techniques, business intelligence tools, and AI-based analytics. This will enable effective evaluation of faculty performance, identification of improvement areas, and enhancement of the overall teaching–learning process in Cost Accounting for B.Com students.

1.2 Motivation

The selection of this project is driven by the growing importance of **data-driven decision-making in education** and the need to enhance teaching effectiveness through systematic analysis of student feedback. In the academic environment, especially in commerce and accounting disciplines, evaluating course delivery and learning outcomes is essential for maintaining quality standards. This project provides an opportunity to apply analytical and technological tools to address these practical challenges.

Personal and Professional Significance

- The project is professionally relevant as it integrates accounting education with data analytics and business intelligence tools, which are highly valued in the fields of finance and commerce.
- It enhances practical skills in data preprocessing, visualization, and AI-assisted analysis, which are essential competencies in modern business environments.
- It provides exposure to tools such as Microsoft Excel, Power BI, and AI-based systems, thereby strengthening analytical and technical capabilities.

Real-World Applications

- Educational institutions can use this model to evaluate faculty performance and course effectiveness.
- Helps in curriculum improvement and academic planning based on data insights.
- Can be extended to other domains such as employee feedback analysis, customer satisfaction studies, and service quality assessment.
- Supports the development of automated reporting systems for institutional decision-making.

Impact on Organization

- Enables institutions to adopt a structured and transparent feedback evaluation system.
- Improves the quality of teaching by identifying strengths and areas for improvement.
- Facilitates evidence-based decision-making in academic administration.
- Enhances student satisfaction and learning outcomes through continuous improvement mechanisms.

Alignment with Modern Industry Trends

- The project aligns with the increasing adoption of Artificial Intelligence and Business Intelligence tools in data analysis.
- Reflects the trend of integrating AI-assisted analytics -Copilot, chat bots into routine data processing tasks.
- Supports the shift towards automation, real-time dashboards, and predictive insights in organizational decision-making.

This project is therefore motivated by its strong academic relevance, practical applicability, and alignment with current technological advancements in AI and data analytics.

1.3 Objectives

- To evaluate student feedback systematically across B.Com specializations for the Cost Accounting course using structured data analytics techniques.
- To identify key performance indicators (KPIs) related to teaching effectiveness, including syllabus coverage, communication, assessment, and student engagement.
- To preprocess and standardize raw feedback data by addressing inconsistencies such as missing values, duplicate records, and invalid responses.
- To analyze faculty performance based on quantitative feedback and compare effectiveness across different specializations.
- To assess the impact of teaching methodologies on student satisfaction and learning outcomes.
- To develop interactive dashboards using Microsoft Excel and Power BI for clear visualization of insights.
- To integrate AI tools -Microsoft Copilot and chatbot systems for automated data analysis, insight generation, and reporting.
- To identify critical improvement areas in course delivery, assessment methods, and student support systems.
- To support data-driven decision-making for academic planning and quality enhancement in higher education.
- To demonstrate the application of business intelligence and AI tools in educational analytics within the accounting domain

1.4 Scope of the Project

- **Dataset Coverage:**
  This project covers student feedback data collected from **480 B.Com students** across four specializations—General, Accounting & Finance, Computer Applications, and Corporate Secretaryship. The dataset includes responses measured using a Likert scale across multiple themes such as syllabus coverage, communication, assessment, engagement, growth opportunities, teaching support, and employability skills.

- **Tools Used:**
  The tools used in this project are limited to Microsoft Excel, Microsoft Power BI, Microsoft Copilot, and AI Chatbot systems for data cleaning, analysis, visualization, and AI-assisted insight generation.

- **Focus of Analysis:**
  The project focuses on data preprocessing (handling missing values, duplicates, and invalid entries), statistical analysis, visualization through dashboards, and AI-assisted reporting. It emphasizes evaluating faculty performance and course effectiveness using structured feedback data.

- **Out of Scope:**
  The project does not include real-time data integration, advanced machine learning or predictive modeling, large-scale database systems, or deployment of a live application environment.

- **Project Nature:**
  This project is intended as an academic and fellowship-based demonstration of applying data analytics and AI tools in education. It is designed for learning and research purposes and is not a production-level or enterprise implementation.

CHAPTER 2
LITERATURE SURVEY.

Research in the domain of student feedback analytics and learning analytics emphasizes the use of data to improve teaching effectiveness, student engagement, and academic performance. Studies indicate that feedback plays a crucial role in enhancing learning outcomes, and integrating analytics enables timely, personalized, and actionable feedback mechanisms

LEARNING ANALYTICS & LEARNING DESIGN

- KATERINA MANGAROSKA &MICHAIL GIANNAKOS

The study highlighted that it enables a shift from traditional teaching methods to personalized and adaptive learning systems. Institutions increasingly use analytics to monitor student performance Improve retention rates and Support academic decision-making. Visualization dashboards are the most widely used tools. Learning analytics improves quality and timeliness of feedback and it supports self-regulated learning and student engagement**.**

STUDENT FEEDBACK & PEER ASSESSMENT ANALYTICS

-JOHN HATTIE & HELEN TIMPERLEY

Highlighted importance of formative assessment and feedback effectiveness Analytical techniques such as: Clustering Predictive analytics Feedback classification were applied. Automated classification of feedback improves efficiency Predictive models help identify student learning gaps. The visualization aids instructors in focusing on critical feedback areas

EDUCATION AND PEDAGOGY –HATTIE& WILIAM

The Study has significantly influenced the understanding of student feedback and assessment practices..Hattie's meta-analytical studies highlight that effective feedback helps learners understand their current performance, identify gaps, and take steps for improvement. Similarly, Wiliam's research on formative assessment underscores the importance of continuous feedback in enhancing classroom learning and student engagement. Both scholars advocate for feedback systems that promote self-regulated learning, where students actively reflect on their progress and take responsibility for their learning outcomes. However, traditional feedback mechanisms are often limited by delayed responses and lack of personalization.

2.2 Tools and Technologies Review

- **Microsoft Excel**

  Microsoft Excel is widely used for data cleaning, transformation, and preliminary analysis. In this project, it is applied for handling missing values, removing duplicates, and standardizing Likert scale responses. Advanced features such as pivot tables, conditional formatting, data validation, and basic statistical functions are used to summarize and organize the dataset efficiently.

- **Microsoft Power BI**

  Microsoft Power BI is employed for developing interactive dashboards and visualizations. It enables the transformation of processed data into meaningful insights through charts, graphs, and reports. Its user-friendly interface allows for the creation of dynamic dashboards that facilitate comparison of faculty performance, course effectiveness, and theme-wise analysis.

- **Microsoft Copilot**

  Microsoft Copilot serves as an AI-powered assistant integrated within Microsoft 365 applications. It is used to automate data analysis tasks, generate summaries, and provide insights using natural language prompts. Copilot enhances productivity by reducing manual effort and enabling quick interpretation of complex datasets.

- **AI chat bots**

  AI chat bots are utilized for generating analytical insights, summarizing findings, and assisting in report preparation. These tools support prompt-based interaction, allowing users to query the dataset and obtain instant, structured responses. They improve efficiency in documentation and interpretation of results.

- **Microsoft Power Query**

  Power Query is used within Excel and Power BI for advanced data transformation and automation of data cleaning processes. It enables efficient handling of large datasets by performing tasks such as filtering, merging, and reshaping data, thereby improving data quality and consistency.

  Overall, the integration of these tools provides a comprehensive framework for data preprocessing, visualization, and AI-assisted analytics, ensuring accurate and efficient evaluation of student feedback in the Cost Accounting course.

2.3 Gaps Identified

- **Lack of Structured Feedback Datasets:**
  Many studies rely on unstructured or limited datasets, with insufficient emphasis on well-organized, theme-based feedback data suitable for systematic analysis.
- **Overdependence on Complex Models:**
  Existing research frequently uses advanced machine learning and statistical models, which are not easily accessible or implementable by non-technical academic users.
- **Limited Practical Implementation:**
  There is a gap between theoretical research and practical application, particularly in using simple tools like Excel and Power BI for real-world academic feedback analysis.
- **Insufficient Integration of AI Tools:**
  Minimal research is available on integrating AI tools such as Microsoft Copilot and AI chatbots with traditional data analysis platforms for educational feedback systems.
- **Data Quality Issues Not Addressed:**
  Many studies do not adequately focus on pr processing challenges such as missing values, duplicates, and inconsistent Likert scale responses.
- **Lack of Real-Time and Automated Insights:**
  Traditional feedback systems lack automation and do not provide real-time or prompt-based analytical insights for decision-making.
- **Limited Focus on Faculty Performance Evaluation:**
  Existing literature emphasizes student performance more than structured evaluation of faculty teaching effectiveness using feedback data.
- **Inadequate Visualization Techniques:**
  Many approaches fail to utilize interactive dashboards and business intelligence tools for clear and user-friendly representation of insights.
- **Absence of Standardized Analytical Framework:**
  There is no widely accepted framework combining data cleaning, analysis, visualization, and AI integration specifically for student feedback analytics.
- **Restricted Domain-Specific Applications:**
  Limited studies focus specifically on commerce and accounting education, such as Cost Accounting, creating a gap in domain-specific analytical models.

CHAPTER 3
PROPOSED METHODOLOGY

3.1 SYSTEM DESIGN

The system design of this project follows a structured data analytics framework, where raw student feedback data is transformed into meaningful insights through systematic stages of data processing, analysis, and visualization. The overall design ensures a smooth flow of data from input to output, enabling effective evaluation of teaching and learning outcomes.

1. Data Source / Input

The raw dataset comprises responses collected from students for the Cost Accounting course. The dataset is organized in tabular format combining students real time responses and data set source from kaggle. This raw dataset forms the foundation for computing composite indices, performing comparative analysis across faculty, and generating interactive dashboards for academic performance evaluation.

Table-1-Raw data set summary-Edu Pulse-Student Feedback on Smart Learning

Course	Bachelor of Commerce
Specialization	B.Com General B.Com Accounting & Finance B.Com Computer Applications B.Com Corporate Secretaryship
Subject	Cost Accounting
Faculty Name	Prof. Meena, Prof. Kumar, Prof. Seetha, Prof. Suresh
Total Student Responses	500
Actual Responses	480
Feedback Variables	Syllabus Coverage and Preparation, Communication and Methodology, Assessment and Feedback, Growth and Opportunities, Teaching Support and Engagement Skills and Employability
Analytics	Likert Scale Rating Scores

The Structured data source for this project is a student feedback dataset collected from **480 B.Com students enrolled in the Cost Accounting course across four specializations:**

- B.Com General (A & B)
- B.Com Accounting & Finance (A & B)
- B.Com Computer Applications (A & B)
- B.Com Corporate Secretaryship (A & B)

The dataset is stored in **Excel (.xlsx) format** and consists of:

- Student identifiers and course details
- Faculty information -4 faculty members handling different specializations

Table-2-Cleaned data set summary-Edu Pulse-Student Feedback on Smart Learning

Faculty Name	B.Com-Specialization	Total Students Enrolled
Prof. Meena	B.Com-General	120
Prof. Kumar	B.Com-Corporate Secretary ship	120
Prof. Seetha	B.Com-Accounting and Finance	120
Prof. Suresh	B.Com-Computer Application	120
	Total Responses	**480**

Primary Feedback Variables-Theme-based Feedback Variables

- Syllabus Coverage and Preparation -SC
- Communication and Methodology -CM
- Assessment and Feedback -AF
- Growth and Opportunities -GO
- Teaching Support and Engagement -TE
- Skills and Employability –SE
- Student Fulfillment_SF

Secondary Feedback Variables- Subthemes-based Feedback Variables

Theme based Feedback Variables (Primary Feedback Variables)	Subthemes based Feedback Variables (Secondary Feedback Variables)
Syllabus Coverage and Preparation (SC)	**SC_Q1:** Defined Syllabus clarity **SC_Q2:** Planned Schedule adherence **SC_Q3:** Adequate Topic explanation **SC_Q4:** Sufficient Lecture preparation **SC_Q5:** Structured learning Framework

Communication and Methodology (CM)	**CM_Q1:** Concepts clarity **CM_Q2:** Appropriate Teaching Method **CM_Q3:** Real time Examples and Illustrations **CM_Q4:** Good Teaching Pace **CM_Q5:** Immediate Query Resolution
Assessment and Feedback (AF)	**AF_Q1:** Regular Assessment **AF_Q2:** Exam Oriented Revision **AF_Q3:** Timely Feedback **AF_Q4:** Monitor Performance Status **AF_Q5:** Evaluation Transparency
Growth and Opportunities (GO)	**GO_Q1:** Analytical Skills **GO_Q2:** Practical Application **GO_Q3:** Academic Growth **GO_Q4:** Student Participation **GO_Q5:** Career Growth
Teaching Support and Engagement (TE)	**TE_Q1:** Academic Support **TE_Q2:** Study Material Quality **TE_Q3:** Digital tools Usage **TE_Q4:** Class Engagement **TE_Q5:** Student Motivation
Skills and Employability (SE)	**SE_Q1:** Practical Skills **SE_Q2:** Industry Relevance **SE_Q3:** Decision Making **SE_Q4:** Employment Utility **SE_Q5:** Professional Foundation

Key Performance Variables:

Key Performance Variables	Subthemes based Feedback Variables
Student Fulfillment **About the Faculty and Course**	**SF_Q1:** Teaching Quality Satisfaction **SF_Q2:** Learning Expectations **SF_Q3:** Faculty Recommendation Score **SF_Q4:** Learning Growth Environment **SF_Q5:** Course Recommendation Score
Improvement Area Indicators	**IA_Q1:** Practical Exposure **IA_Q2:** Resource adequacy **IA_Q3:** Revision Support **IA_Q4:** Teaching Enhancement **IA_Q5:** Applied Assessment

Faculty Overall Rating-Performance Metrics
Overall Rating: Rate the overall performance of the faculty.
(1 – Poor | 2 – Fair | 3 – Good | 4 – Very Good | 5 – Excellent)

Likert Scale:
1 – Strongly Disagree | 2 – Disagree | 3 – Neutral | 4 – Agree | 5 – Strongly Agree

The Analysis employs a **5-point Likert scale** ranging from 1 (Strongly Disagree) to 5 (Strongly Agree) to quantify student perceptions regarding subject delivery and faculty performance in Cost Accounting. Each response is assigned a numerical weight, enabling the transformation of qualitative opinions into measurable data. The collected responses are aggregated to compute Average scores, satisfaction indices, and comparative performance metrics across different dimensions such as syllabus coverage, teaching effectiveness, assessment, and employability. Higher Average scores indicate stronger agreement and higher satisfaction levels, while lower scores highlight areas requiring improvement. This scaling technique facilitates **benchmarking, and dashboard visualization**, thereby supporting objective evaluation and data-driven academic decision-making.

2. Data Cleaning

Removed Duplicates

- Checked for repeated Student_ID entries.
- Ensured each row represented a unique student response.
- Dropped any exact duplicates to maintain data integrity.

Handled Null or Missing Values

- Scanned all columns for blanks or missing entries.
- For numerical feedback questions named SE_Q1 to SE_Q5, replaced missing values with the **Average score** of that question to preserve statistical balance.

Standardized Formats

- Unified rating scales to a consistent 1–5 range.
- Trimmed extra spaces and corrected inconsistent naming.
- Ensured column headers were clean and descriptive for dashboard mapping

3. Analysis & Processing Techniques

❖ Excel-Based Analysis

- **Pivot Tables**:
  - Created pivot tables to summarize average scores per faculty, course, Theme based and sub theme based feedback variable
  - Used Filters to compare Performance across Faculty members
- **Formulas Used**:
  - AVERAGEIFS to calculate theme-wise scores by faculty/course.
  - COUNTIFS to track rating distributions- how many students gave a score of 5

❖ Power BI Dashboard

- **Data Model Setup**:
  - Imported cleaned Excel data into Power BI.
  - Created relationships between Student_ID, Course, and Faculty for dynamic filtering.

- **Measures Created**:

Teaching Quality

Teaching_Quality = DIVIDE('Table'[SC_Avg] + 'Table'[CM_Avg] + 'Table'[AF_Avg], 3)

Learning Outcome

Learning_Outcome = DIVIDE('Table'[GO_Avg] + 'Table'[SE_Avg], 2)

Scholastic Experience

Scholastic_Experience = DIVIDE('Table'[TE_Avg] + 'Table'[SF_Avg], 2)

❖ **AI Prompting & Insight Extraction- Prompt Design Structure**

- **A. Comparative Analysis Prompts-**"Compare Teaching Excellence Scores across faculties - highlight performance gaps."
- **B. Diagnostic Insight Prompts-** "Analyze the Scores of the Learning outcomes of the students based on Faculty Performance Metrics"
- **C. Predictive / Prescriptive Prompts-**"Suggest improvement strategies for faculties with lower satisfaction scores."
- **Descriptive Analysis Prompts-**"Summarize the overall student satisfaction levels across all faculties."

4. Visualization and Output

Faculty-wise performance comparison Course-wise satisfaction levels Theme-wise analysis give dashboard visualizations with insights. The structured dashboard-style analysis of the **Cost Accounting Feedback Dataset** with faculty-wise, course-wise, and theme-wise insights.

Dash Board Components

Table-3 KEY PERFORMANCE INDICATORS-COMPARATIVE ANALYTICS

1.	Teaching Excellence Pulse-Teaching Performance Comparative Analytics
2.	Learning Outcome Excellence Pulse-Comparative Analytics
3.	Scholastics Excellence Pulse- Comparative Analytics
4.	Pulse Point-Faculty Performance Ranking Board

Dashboard Visualization and Insights

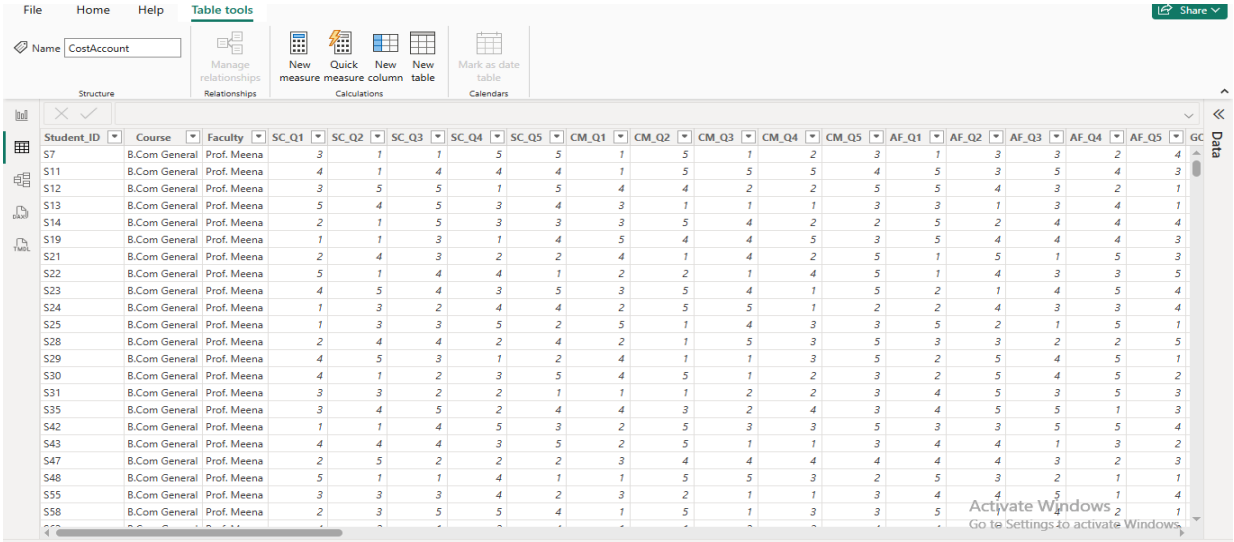
1. **Clustered Column Chart** – Average ratings across SE, SC, CM, AF, GO for all four faculties
2. **Line or Area chart** – Syllabus coverage pulse and Assessment and Feedback Pulse Analytics of the faculty members.
3. **Trend Line** – Trend line for Teaching support and Student Engagement
4. **Bar graph**-Faculty performance Matrix for Student Fulfillment of the course Cost accounting.

3.2 MODULES / COMPONENTS USED

MODULE 1 — DATA INGESTION

- Imported the raw dataset into **Excel** for initial review.
- Loaded into **Power BI** for dynamic analysis and visualization.
- Ensured column headers and data types were preserved during import

FIGURE-1 RAW/UNCLEANED DATASET-EDUPULSE



Student_ID	Course	Faculty	SC_Q1	SC_Q2	SC_Q3	SC_Q4	SC_Q5	CM_Q1	CM_Q2	CM_Q3	CM_Q4	CM_Q5	AF_Q1	AF_Q2	AF_Q3	AF_Q4	AF_Q5	GC
S7	B.Com General	Prof. Meena	3	1	1	5	5	1	5	1	2	3	1	3	3	2	4	
S11	B.Com General	Prof. Meena	4	1	4	4	4	1	5	5	5	4	5	3	5	4	3	
S12	B.Com General	Prof. Meena	3	5	5	1	5	4	4	2	2	5	5	4	3	2	1	
S13	B.Com General	Prof. Meena	5	4	5	3	4	3	1	1	1	3	3	1	3	4	1	
S14	B.Com General	Prof. Meena	2	1	5	3	3	3	5	4	2	2	5	2	4	4	4	
S19	B.Com General	Prof. Meena	1	1	3	1	4	5	4	4	5	3	5	4	4	4	3	
S21	B.Com General	Prof. Meena	2	4	3	2	2	4	1	4	2	5	1	5	1	5	3	
S22	B.Com General	Prof. Meena	5	1	4	4	1	2	2	1	4	5	1	4	3	3	5	
S23	B.Com General	Prof. Meena	4	5	4	3	5	3	5	4	1	5	2	1	4	5	4	
S24	B.Com General	Prof. Meena	1	3	2	4	4	2	5	5	1	2	2	4	3	3	4	
S25	B.Com General	Prof. Meena	1	3	3	5	2	5	1	4	3	3	5	2	1	5	1	
S28	B.Com General	Prof. Meena	2	4	4	2	4	2	1	5	3	5	3	3	2	2	5	
S29	B.Com General	Prof. Meena	4	5	3	1	2	4	1	1	3	5	2	5	4	5	1	
S30	B.Com General	Prof. Meena	4	1	2	3	5	4	5	1	2	3	2	5	4	5	2	
S31	B.Com General	Prof. Meena	3	3	2	2	1	1	1	2	2	3	4	5	3	5	3	
S35	B.Com General	Prof. Meena	3	4	5	2	4	4	3	2	4	3	4	5	5	1	3	
S42	B.Com General	Prof. Meena	1	1	4	5	3	2	5	3	3	5	3	3	5	5	4	
S43	B.Com General	Prof. Meena	4	4	4	3	5	2	5	1	1	3	4	4	1	3	2	
S47	B.Com General	Prof. Meena	2	5	2	2	2	3	4	4	4	4	4	4	3	2	3	
S48	B.Com General	Prof. Meena	5	1	1	4	1	1	5	5	3	2	5	3	2	1	1	
S55	B.Com General	Prof. Meena	3	3	3	4	2	3	2	1	1	3	4	4	5	1	4	
S58	B.Com General	Prof. Meena	2	3	5	5	4	1	5	1	3	3	5	[illegible]	[illegible]	2	1	

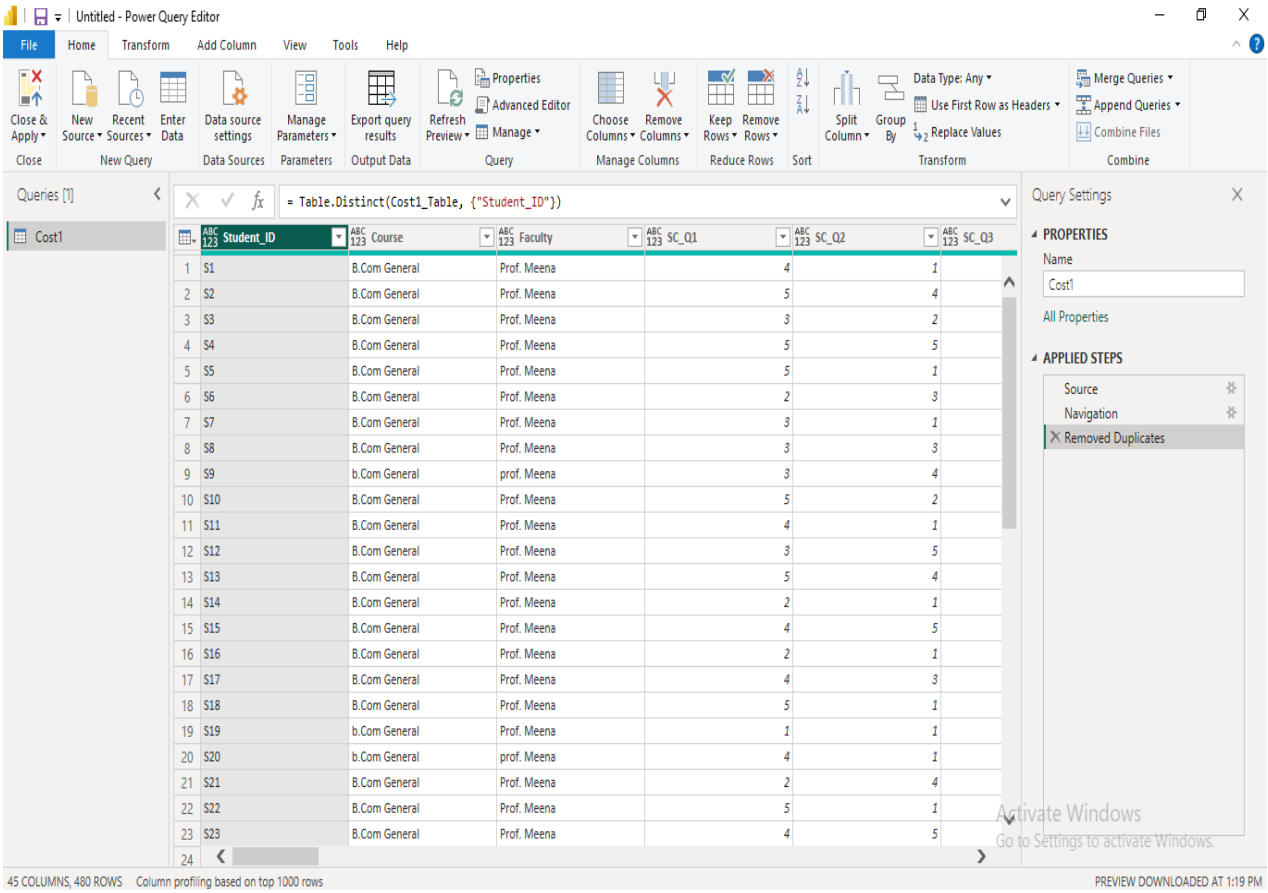
MODULE 2 DATA CLEANING IN POWER BI AND EXCEL

In Microsoft Excel –remove duplicates to get the structured dataset of 480 rows and 44 Columns for faculty performance analytics

In Power-BI

- Removed duplicate Student_ID entries in Microsoft Excel
- Load the Data Open Power BI Desktop.
  - Click on Home in the top ribbon
  - Click Get Data -Choose Excel or Text/CSV
  - Select dataset and click Load-Click Transform Data
  - In Power Query Editor- Check that all Sub themes Feedback Variables columns such as SC Q1, CM Q1, etc. are of type Whole Number
  - Ensure column names are clean and consistent
  - Replace values and Errors
  - Add Additional Columns in creating Average of SC,CM,AF,GO, TE, SE(Q1-

FIGURE 2-CLEANED DATASET-EDUPULSE



	Student_ID	Course	Faculty	SC_Q1	SC_Q2	SC_Q3
1	S1	B.Com General	Prof. Meena	4	1	
2	S2	B.Com General	Prof. Meena	5	4	
3	S3	B.Com General	Prof. Meena	3	2	
4	S4	B.Com General	Prof. Meena	5	5	
5	S5	B.Com General	Prof. Meena	5	1	
6	S6	B.Com General	Prof. Meena	2	3	
7	S7	B.Com General	Prof. Meena	3	1	
8	S8	B.Com General	Prof. Meena	3	3	
9	S9	b.Com General	prof. Meena	3	4	
10	S10	B.Com General	Prof. Meena	5	2	
11	S11	B.Com General	Prof. Meena	4	1	
12	S12	B.Com General	Prof. Meena	3	5	
13	S13	B.Com General	Prof. Meena	5	4	
14	S14	B.Com General	Prof. Meena	2	1	
15	S15	B.Com General	Prof. Meena	4	5	
16	S16	B.Com General	Prof. Meena	2	1	
17	S17	B.Com General	Prof. Meena	4	3	
18	S18	B.Com General	Prof. Meena	5	1	
19	S19	b.Com General	Prof. Meena	1	1	
20	S20	b.Com General	prof. Meena	4	1	
21	S21	B.Com General	Prof. Meena	2	4	
22	S22	B.Com General	Prof. Meena	5	1	
23	S23	B.Com General	Prof. Meena	4	5	

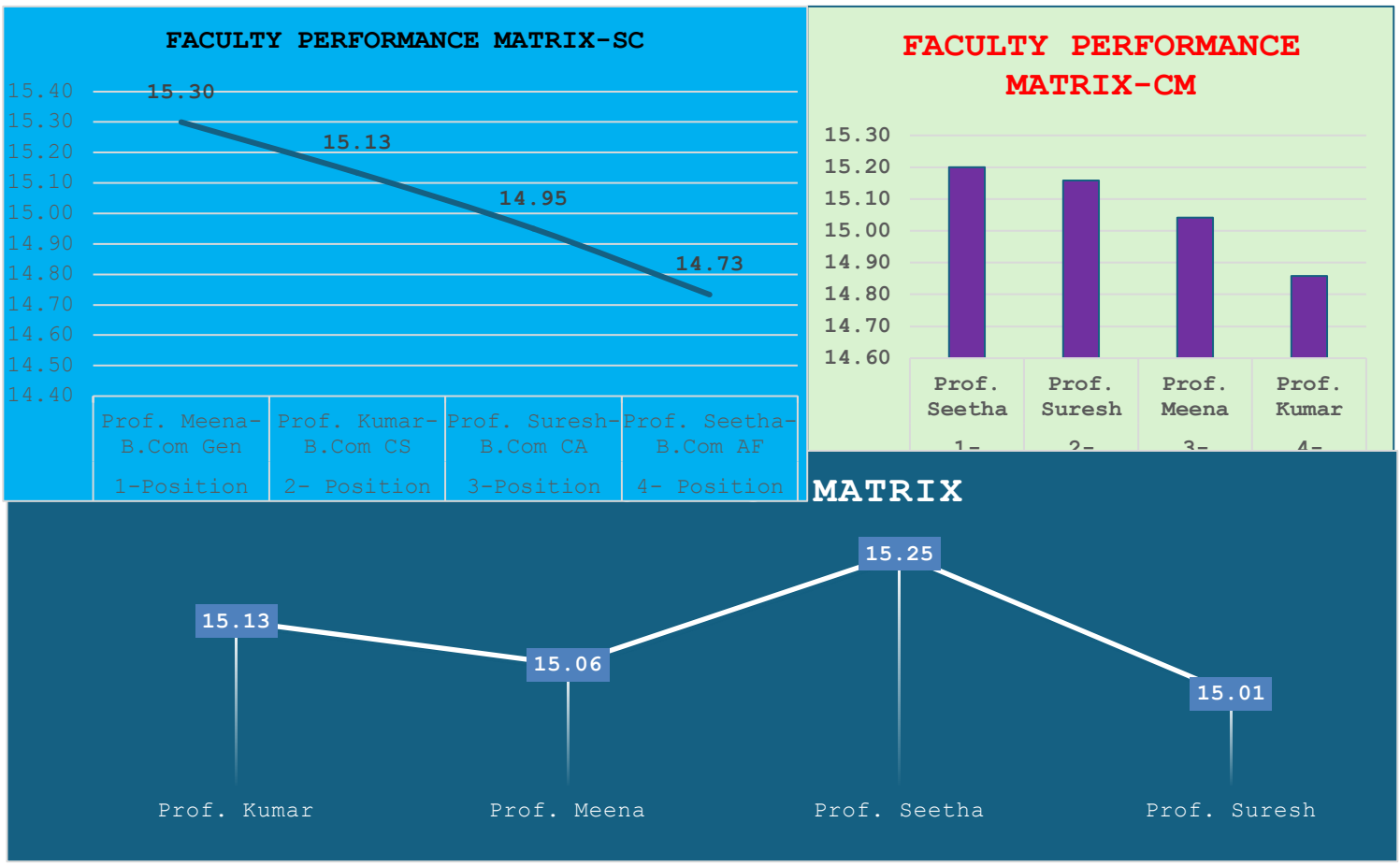
MODULE 3 — DATA ANALYSIS

Multi -Dimensional Feedback Analysis

- **DIMENSION -1 TEACHING EXCELLENCE PULSE**

Teaching excellence pulse is measured by three essential parameters—**SC (Syllabus Coverage and Presentation), CM (Communication and Methodology), and AF (Assessment and Feedback)**. **SC** reflects how thoroughly the syllabus is covered and how effectively it is presented to students, ensuring clarity and engagement. **CM e**mphasizes the faculty's ability to communicate concepts, adopt suitable teaching methodologies, and create an interactive learning environment. **AF** highlights the quality of evaluation, feedback, and academic guidance provided to students, supporting their continuous improvement. Together, these three dimensions form a composite **Teaching Excellence Score**, offering a holistic measure of faculty performance. This integrated approach ensures that teaching excellence is not judged by a single aspect but by the synergy of syllabus mastery, effective communication, and constructive feedback.

Figure-3 TEACHING EXCELLENCE-COMPARATIVE ANALYTICS



➢ DIMENSION -2 LEARNING OUTCOME EXCELLENCE PULSE

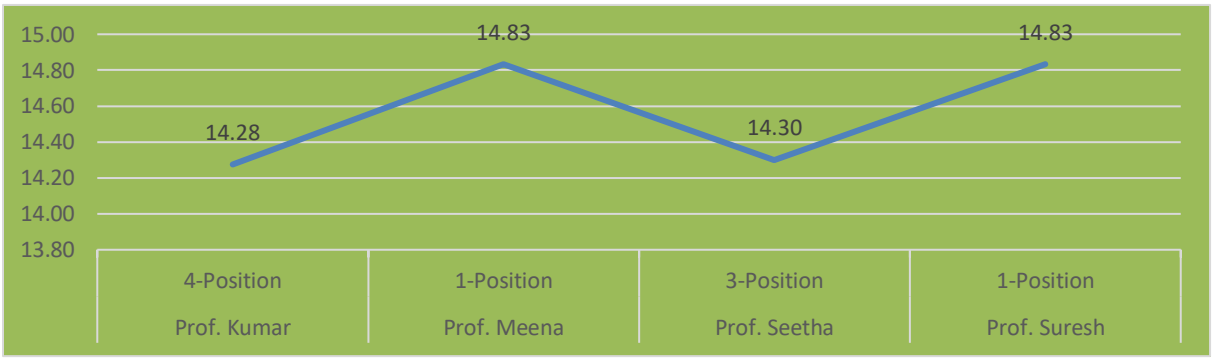
The Learning Outcome Pulse is designed to measure how effectively faculty teaching translates into student development. It integrates two critical dimensions:

GO – Growth & Opportunities- Captures how faculty create pathways for students to expand knowledge, explore new domains, and access opportunities for academic and career progression. High GO scores indicate that students are being exposed to diverse learning experiences, research opportunities, and innovative projects.

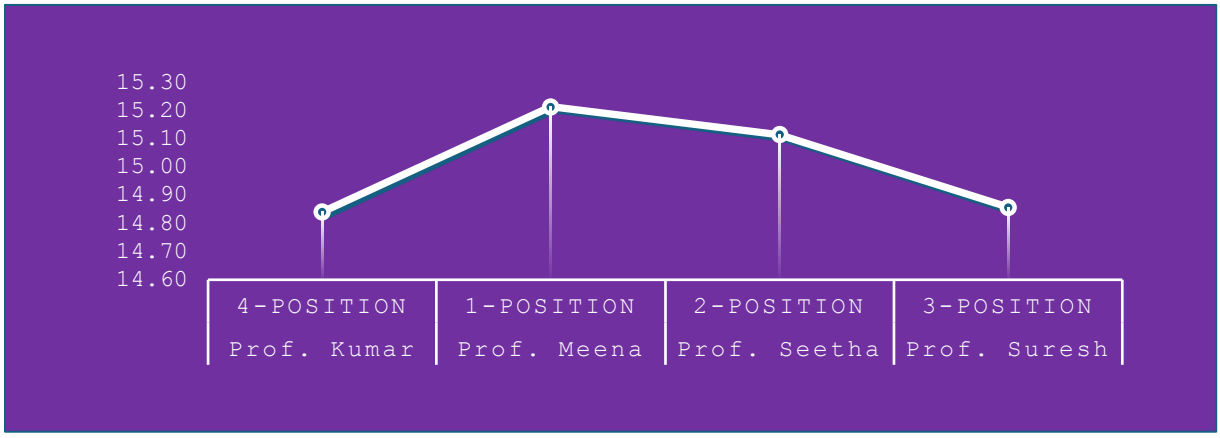
SE – Skills & Employability-Evaluates how teaching equips students with practical skills, industry readiness, and employability competencies. Strong SE scores reflect faculty success in bridging theory with practice, fostering problem-solving, communication, and career-oriented skills. Together, these parameters form the Learning Outcome Pulse Score, which provides a holistic view of how teaching impacts both student growth trajectories and career readiness.

Figure-4 LEARNING OUTCOME EXCELLENCE-COMPARATIVE ANALYTICS

GO – Growth & Opportunities



SE – Skills & Employability



➢ DIMENSION -3 SCHOLASTIC EXCELLENCE PULSE

The Scholastic Excellence Pulse measures how effectively faculty teaching contributes to students' academic growth and satisfaction. It integrates two complementary dimensions:

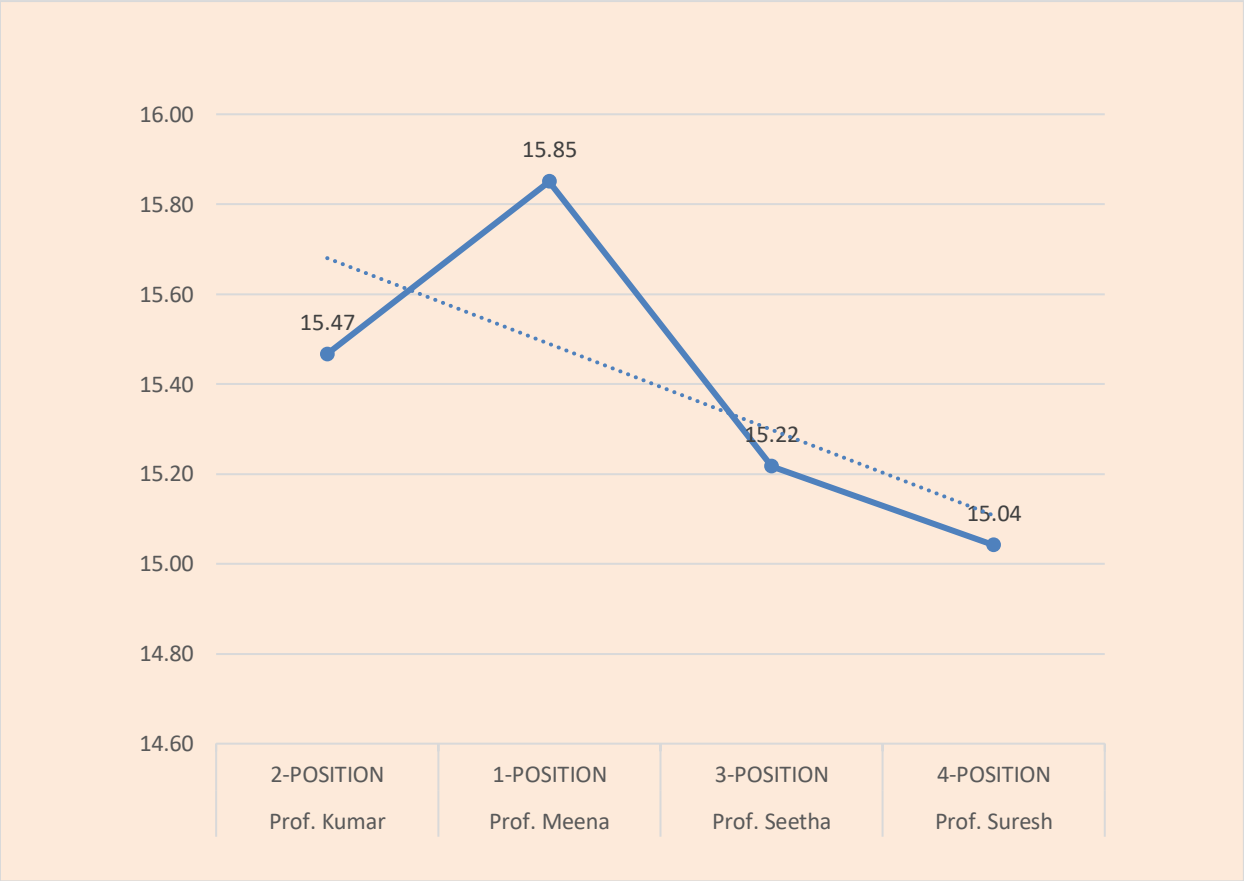
TE – Teacher Support & Student Engagement-Reflects how actively faculty engage students through interactive teaching, mentoring, and academic guidance. High TE scores indicate strong faculty involvement, motivating students to participate and perform better.

SF – Student Fulfillment-Captures how teaching outcomes translate into student satisfaction, confidence, and academic fulfillment. High SF scores show that students feel supported, valued, and satisfied with their learning experience.

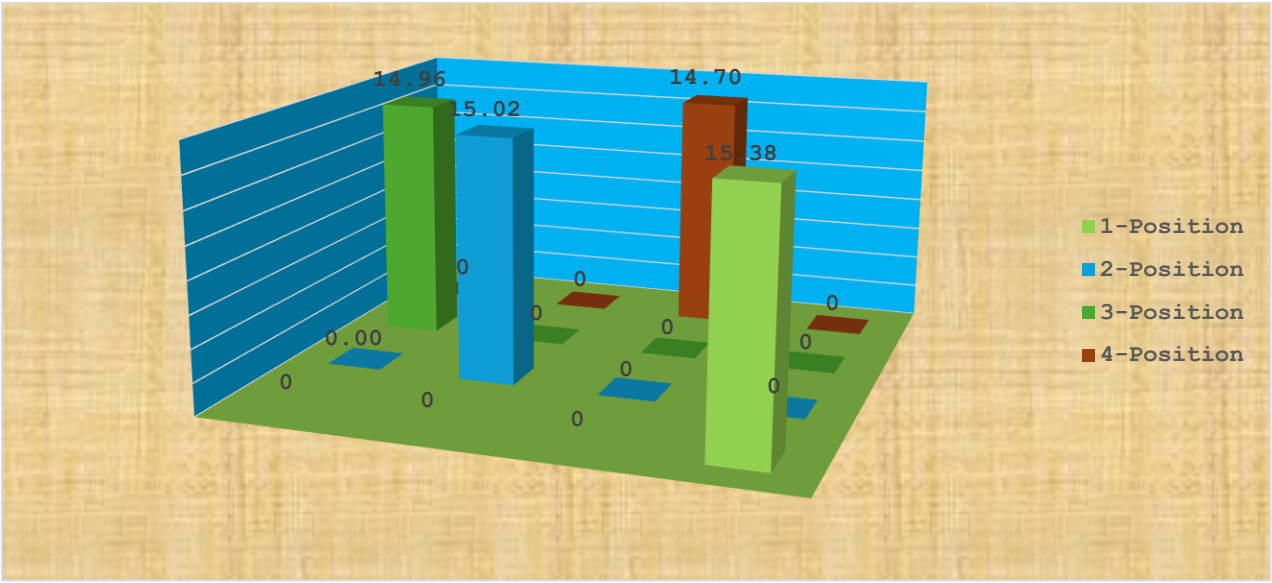
Together, these parameters form the Scholastic Excellence Score, offering a holistic view of how teaching quality and student satisfaction combine to create academic excellence

Figure-5 SCHOLASTIC EXCELLENCE-COMPARATIVE ANALYTICS

TE – TEACHER SUPPORT & STUDENT ENGAGEMENT



SF – STUDENT FULFILLMENT



Student FeebacK-SF Score	Position			
Faculty	1-Position	2-Position	3-Position	4-Position
Prof. Kumar			14.96	
Prof. Meena		15.02		
Prof. Seetha				14.70
Prof. Suresh	15.38			

MODULE 4 — VISUALIZATION / REPORT GENERATION

TABLE -4 TEACHING EXCELLENCE PULSE – COMPARATIVE FACULTY PERFORMANCE METRICS

Faculty	SC – Syllabus Coverage & Presentation	CM – Communication & Methodology	AF – Assessment & Feedback	Teaching Excellence Score	Excellence Position
Prof. Meena	15.30	15.04	15.06	**15.13**	1st Position
Prof. Suresh	14.95	15.16	15.01	**15.04**	2nd Position
Prof. Seetha	14.73	**15.20**	**15.25**	**15.06**	3rd Position
Prof. Kumar	15.13	14.86	15.13	**15.04**	4th Position

TABLE-5- LEARNING OUTCOME EXCELLENCE PULSE – COMPARATIVE FACULTY PERFORMANCE METRICS

Faculty	GO (Growth & Opportunities)	SE (Skills & Employability)	LO Excellence Score	Position	Key Interpretation
Prof. Meena	14.83	15.22	15.03	1st Position	Top performer—creates strong growth pathways while excelling in employability-focused teaching.
Prof. Suresh	14.83	14.86	14.85	2nd Position	Balanced performer—steady and reliable across both parameters.
Prof. Seetha	14.30	15.12	14.71	3rdPosition	Strong in employability skills, but slightly weaker in growth opportunities.
Prof. Kumar	14.28	14.84	14.56	4th Position	Needs improvement—lower scores in both parameters, especially in growth opportunities.

TABLE-6- LEARNING OUTCOME EXCELLENCE PULSE – COMPARATIVE FACULTY PERFORMANCE METRICS

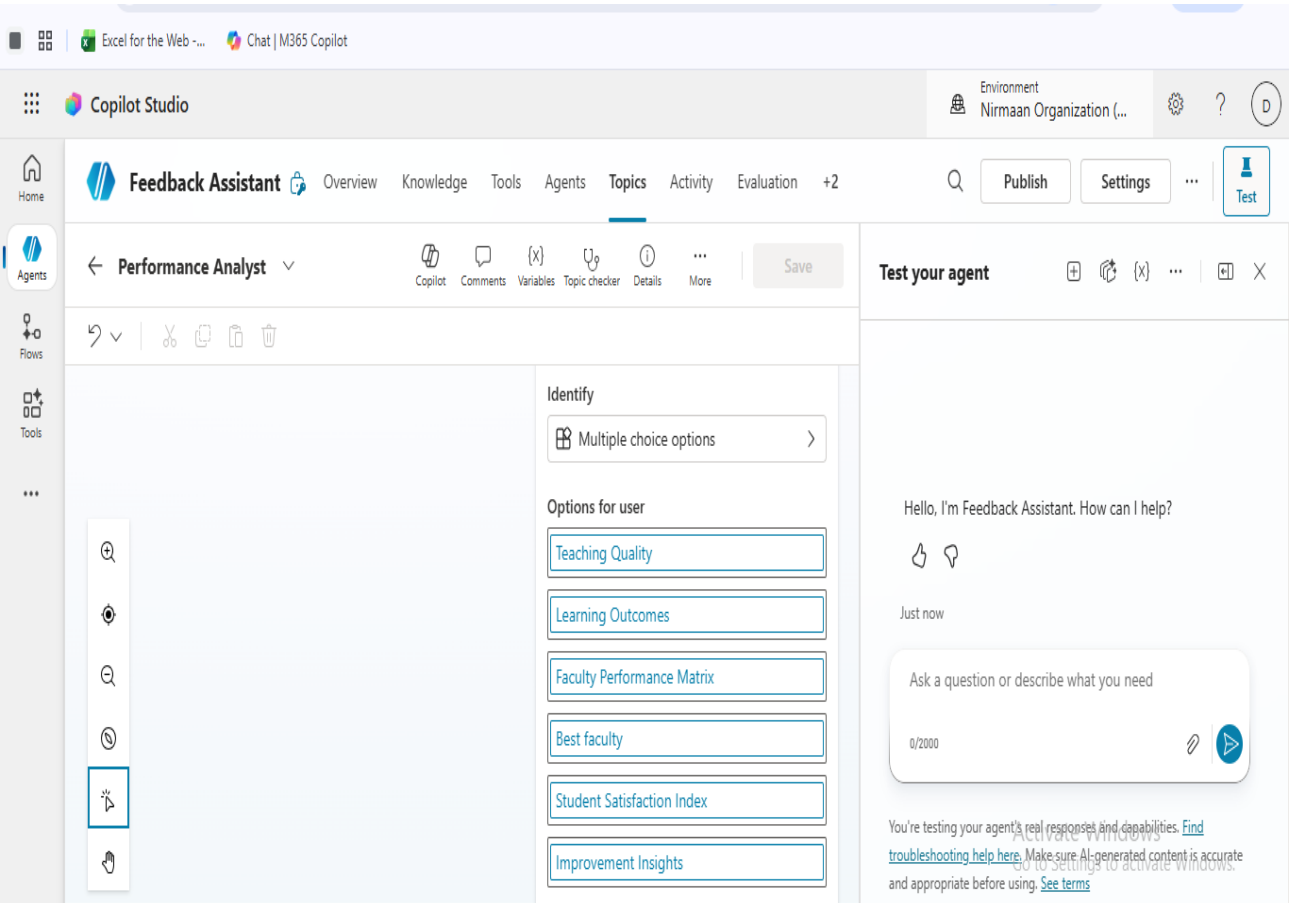
Faculty	TE Score	SF Score	Scholastic Excellence Score	Position	Key Interpretation
Prof. Meena	**15.85**	15.02	**15.44**	1st	Top performer—strong teacher support and high student engagement lead to superior scholastic outcomes.
Prof. Kumar	15.47	14.96	15.22	2nd	Balanced performer—consistent teaching support with moderate student fulfillment.
Prof. Suresh	15.04	**15.38**	15.21	3rd	High student fulfillment—students feel confident and satisfied, though engagement can be strengthened.
Prof. Seetha	15.22	14.70	14.96	4th	Needs improvement—good teaching effort but lower student fulfillment indicates scope for better feedback and support.

MODULE 5 — AI INTEGRATION

AI-TOOL-COPILOT STUDIO- **FEEDBACK ASSISTANT CHATBOT**

Performance Analyst-KPI Outcomes

Figure-6 FEEDBACK ASSISTANT CHATBOT



MODULE 6 — INSIGHT EXTRACTION & RECOMMENDATION

Used **Copilot** to:

- Generate Insights for Faculty Performance Metrics
- Generate Measures For Performance Improvement and Enhancement
- Design Feedback layout
- Draft report narratives and presentation content

TEACHING EXCELLENCE INSIGHTS

- **Balanced Excellence:** Scores are tightly clustered (15.04–15.13), indicating a uniformly high teaching standard.
- **Parameter Leaders:**
  - SC → Prof. Meena (15.30)
  - CM → Prof. Seetha (15.20)
  - AF → Prof. Seetha (15.25)
- **Mentorship Potential:** Prof. Seetha's strengths in CM and AF can be leveraged to uplift peers, while Prof. Meena's syllabus mastery sets a benchmark.
- **Improvement Zone:** Prof. Kumar's CM score (14.86) suggests scope for enhancing classroom engagement strategies.

LEARNING OUTCOME EXCELLENCE INSIGHTS

 GO Leaders: Prof. Meena and Prof. Suresh (14.83 each) → They are effective in fostering growth opportunities.

 SE Leader: Prof. Meena (15.22) → Sets the benchmark for employability-focused teaching.

 Balanced Performer: Prof. Suresh → Reliable across both dimensions, making him a model of consistency.

 Improvement Zone: Prof. Kumar → Needs targeted support in both growth and employability integration.

SCHOLASTIC EXCELLENCE INSIGHTS

• **TE Leader:** Prof. Meena → Exemplary in teacher support and engagement.

• **SF Leader:** Prof. Suresh → Excels in student satisfaction and fulfillment.

• **Balanced Performer:** Prof. Kumar → Maintains steady performance across both parameters.

• **Improvement Zone:** Prof. Seetha → Needs focus on enhancing student fulfillment and feedback mechanisms.

3.3 DATA FLOW DIAGRAM (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data flows through your system. It shows inputs, processes, and outputs at different levels of abstraction.

DFD Level 0 — Context Diagram-This diagram represents the entire system as a single process: **Flow:**

- **Input**: Raw Feedback Dataset (Excel file with student ratings and comments)
- **Process**: Feedback Analysis System
- **Output**: Cleaned Data → Dashboard Visuals → Insightful Reports

DFD Level 1 — Detailed Process Flow-This diagram breaks down the main process into four sub-processes:

Sub-process 1: Data Loading & Validation

- Load Excel dataset into Excel or Power BI.
- Validate column headers, data types, and completeness.
- Store in **Raw Data Repository**.

Sub-process 2: Data Cleaning & Transformation

- Remove duplicates, handle nulls, standardize formats.
- Convert ratings to numeric, categorize faculty/course.
- Store in **Filtered Data Repository**.

Sub-process 3: Analysis & AI Processing

- Apply formulas, pivot tables.
- Use Copilot/AI to extract insights and summarize feedback.
- Store in **Insights & Metrics Repository**.

Sub-process 4: Visualization & Report Generation

- Build dashboards (bar charts, radar charts, pie charts).
- Generate infographic-style reports and summaries.
- Output to **Dashboard Views** and **Report Files**.

FIGURE-7 DFD-LEVEL 0-CONTEXT DIAGRAM

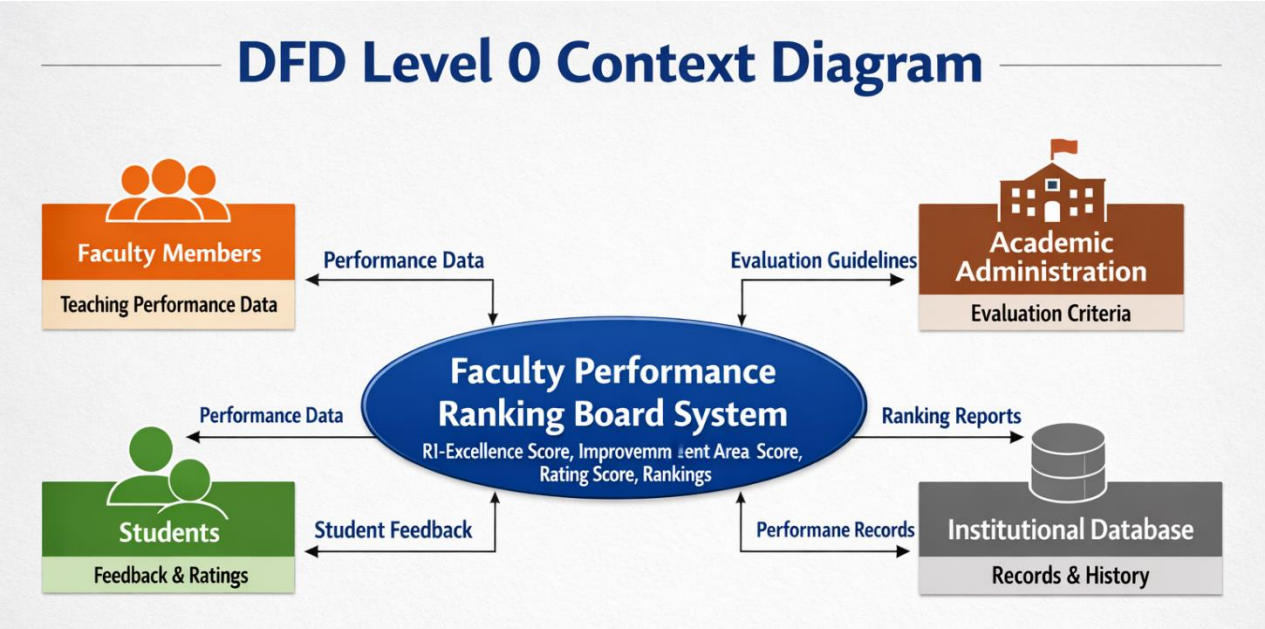
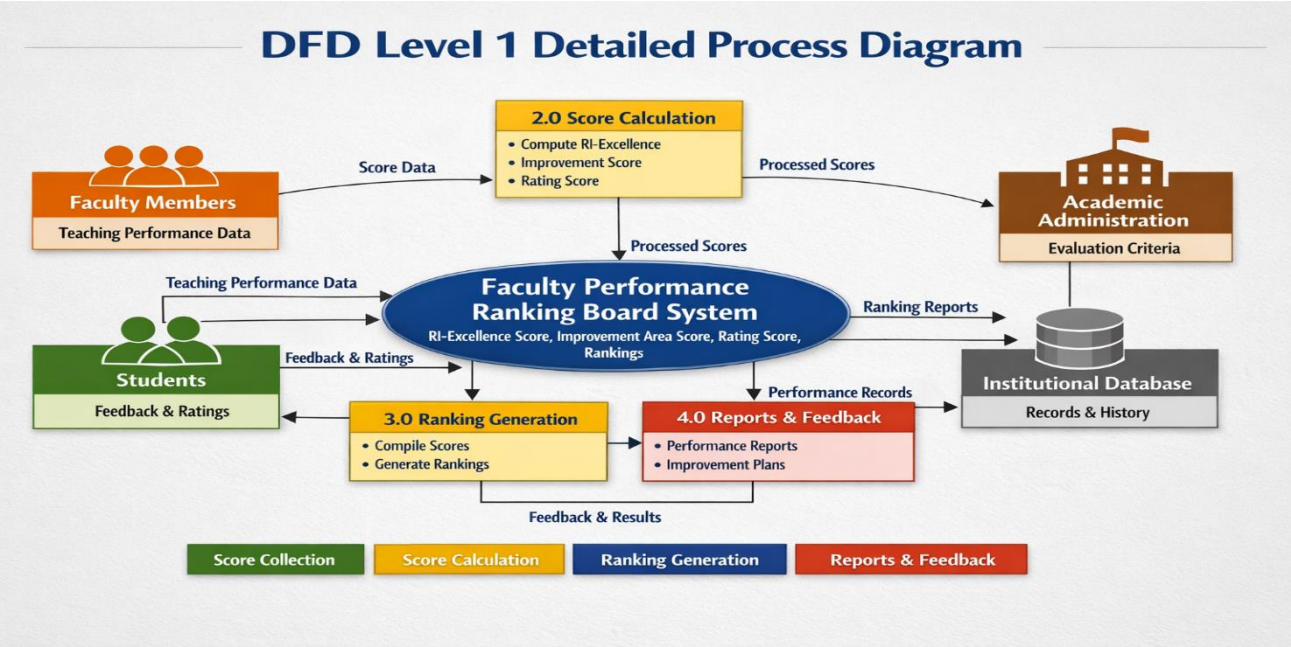
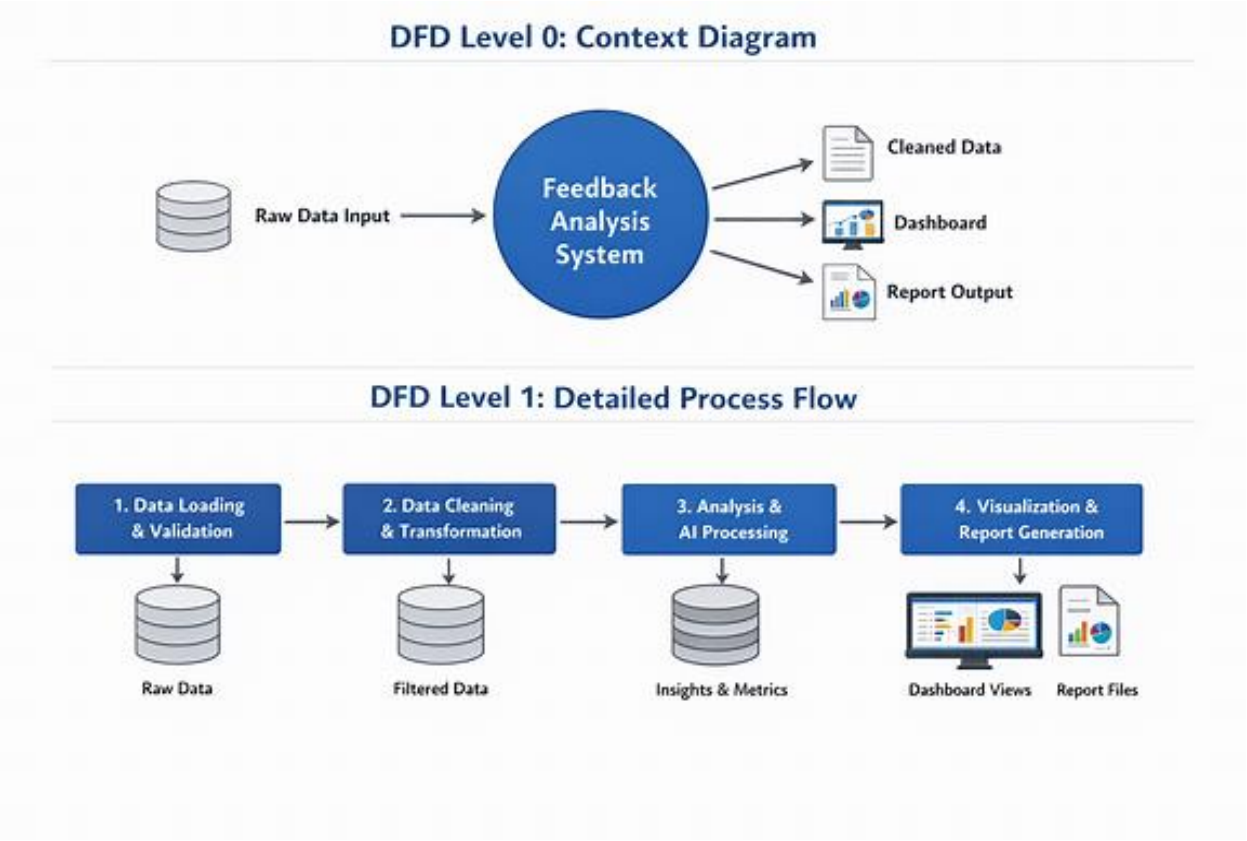


FIGURE-8 DFD-LEVEL 1-DETAILED PROCESS DIAGRAM





3.4 ADVANTAGES

Automated Data Cleaning-Reduces manual errors and ensures consistent formatting, Handles missing values, duplicates, and type mismatches efficiently.

Insightful Analysis with AI Support-Combines traditional formulas (Excel,)with AI-generated summaries, Extracts hidden patterns and recommends actionable improvements.

Interactive Dashboards-Power BI visuals offer dynamic filtering by faculty, course, and theme, Stakeholders can explore data intuitively without technical expertise.

Theme-Based Evaluation-Breaks feedback into meaningful categories (Engagement, Clarity, Materials, Feedback, Outcomes), Enables targeted interventions for each weak area.

Faculty & Course Comparison-Highlights performance gaps between instructors and courses, Supports data-driven decisions for training and curriculum design.

Student-Centric Reporting-Includes student feedback summaries and satisfaction scores, Promotes transparency and responsiveness in academic planning.

Reusability & Adaptability-The framework can be reused for other subjects, semesters, or institutions.**,** Easily integrates with future datasets or survey formats.

3.5 REQUIREMENT SPECIFICATION

TABLE-7 HARDWARE REQUIREMENTS

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

TABLE-8 SOFTWARE REQUIREMENTS

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Power BI Desktop	Latest version (free download from Microsoft)
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	ChatGPT / Google Gemini / Microsoft Copilot (web)
Web Browser	Google Chrome / Microsoft Edge (latest version)

CHAPTER 4
IMPLEMENTATION AND RESULTS

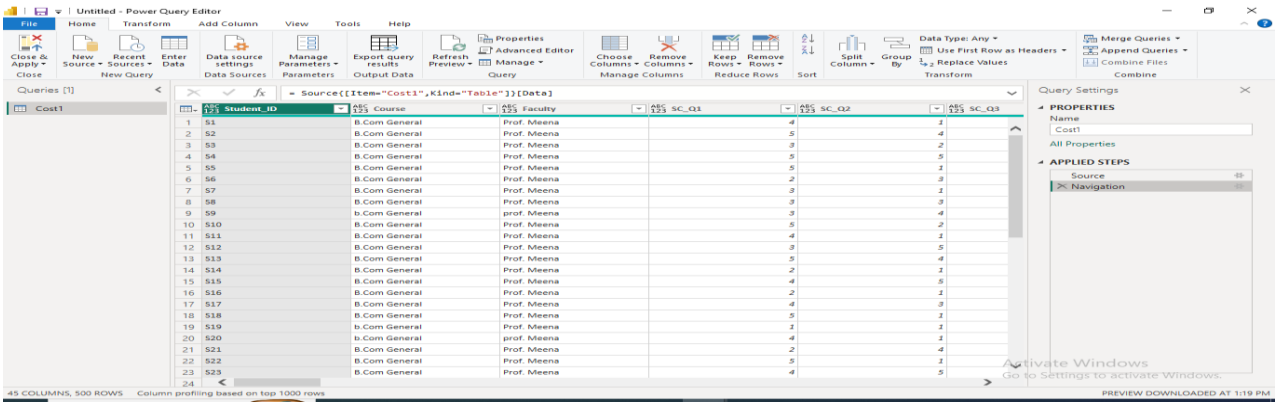
4.1 DATA CLEANING RESULTS

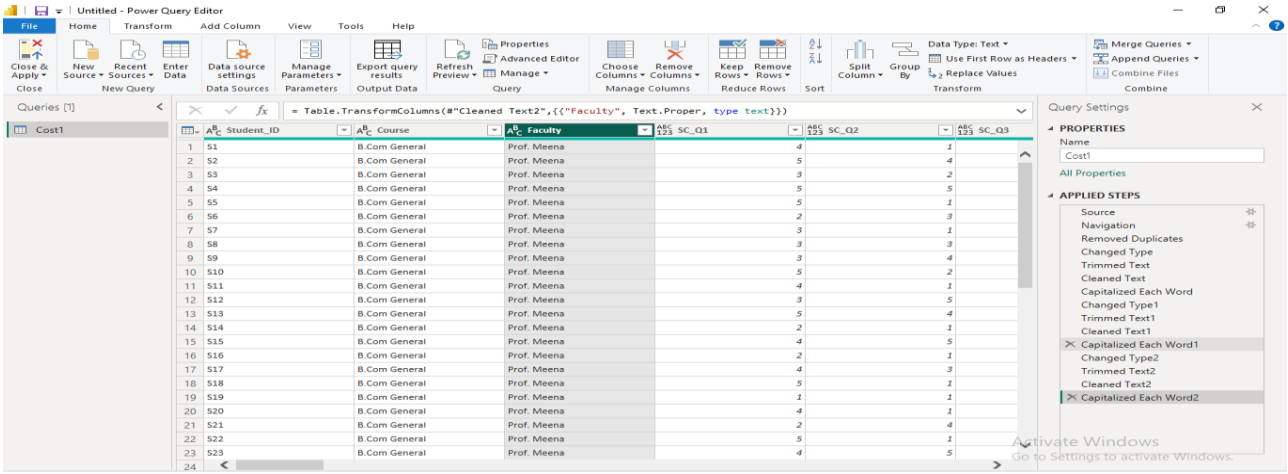
Dataset Source-The dataset base was provided by Kaggle and integrated the feedback collection initiative from the students of our institution for Cost Accounting courses across B.Com General and B.Com AF streams and integrated both and framed Structured Data set.

Raw and Uncleaned dataset size: Number of rows -500 and columns in the raw data-44

Cleaned Dataset Size Number of rows -480 and columns in the raw data-44

FIGURE-9 DATASET CLEANING RESULTS





4.2 VISUALIZATION / OUTPUT RESULTS

MICROSOFT EXCEL –KPI RESULTS

Prof. Meena is the Excellence Benchmark, setting standards for teaching delivery and improvement.

Prof. Suresh demonstrates strong student satisfaction, making him a reliable performer.

Prof. Seetha shows adaptability and growth potential, aligning well with institutional goals.

Prof. Kumar needs structured mentoring, especially in improvement and student engagement

Institutional Decision

- Recognize and reward top performers.
- Use peer-led mentoring (Meena for excellence, Suresh for student satisfaction) to uplift others.
- Provide targeted support for improvement zones.

- Position the Pulse Point Ranking Board as a recurring evaluation tool to sustain excellence and track growth.

FIGURE-10- PULSE POINT FACULTY PERFORMANCERANKING BOARD

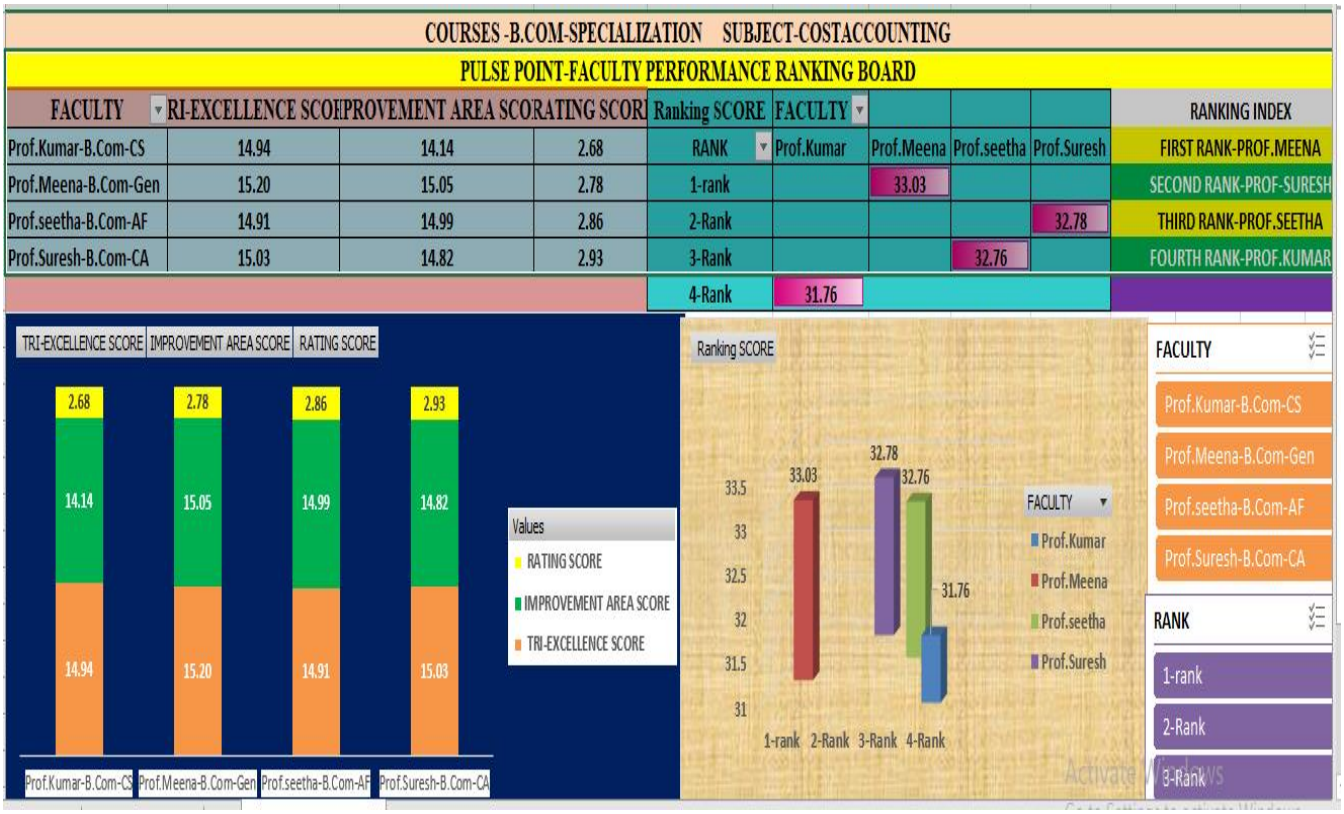
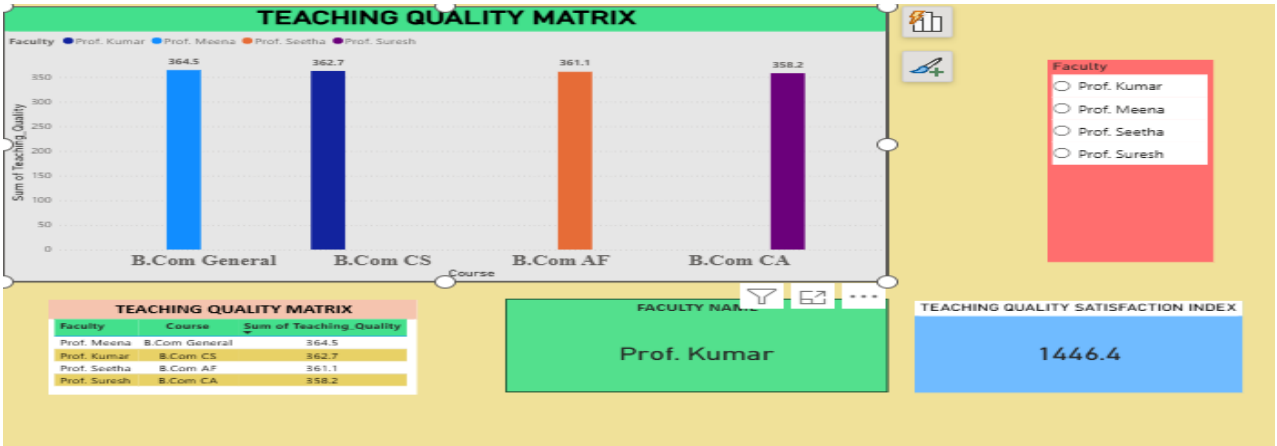
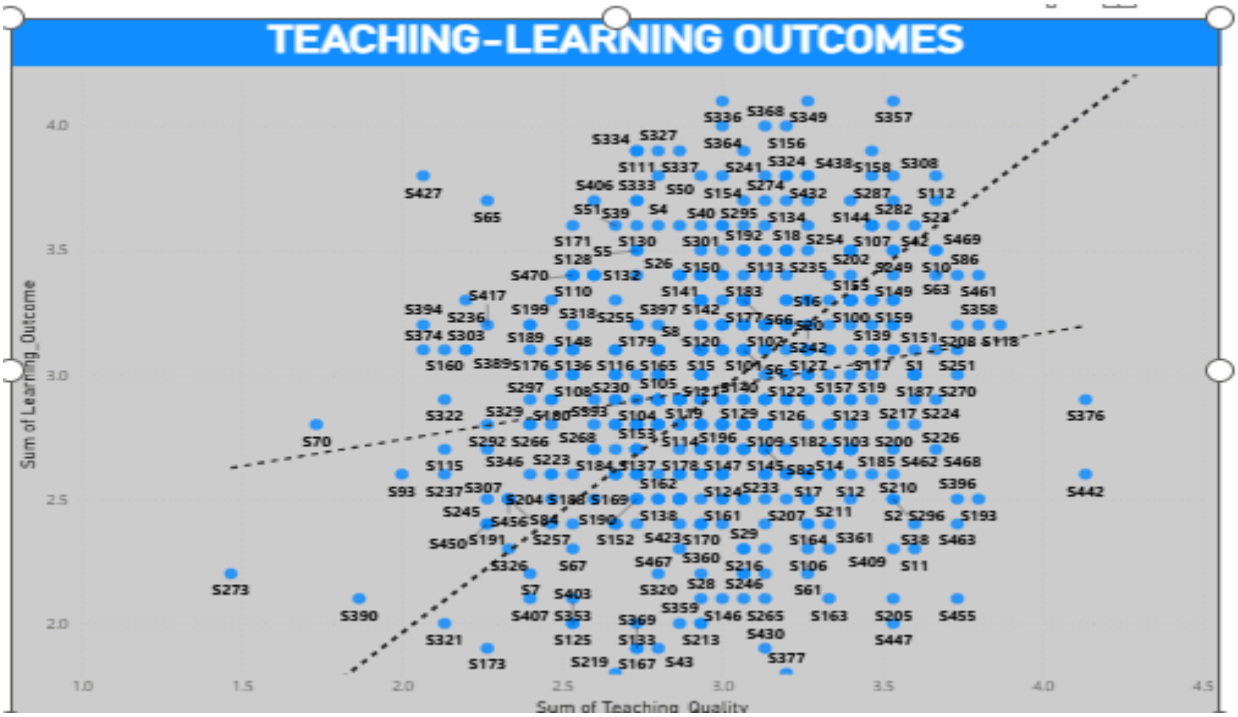


FIGURE-11 POWER BI-DASHBOAD RESULTS

— Faculty Teaching Quality Matrix



Dashboard / Chart 2 — Teaching Learning Outcomes



4.3 AI TOOL INTEGRATION RESULTS

Tool Used-Microsoft Copilot was used for data analysis, visualization design, and insights.

Prompts Used-Some of the key prompts applied during the project included:

1. "Faculty-wise performance comparison, course-wise satisfaction levels, theme-wise analysis — give dashboard visualizations with insights." → Produced comparative dashboards highlighting strengths and weaknesses of each faculty and course.
2. "Give pie chart visuals for overall rating distribution." → Generated clear pie charts showing how students rated faculty overall (1–5 scale).

Output Quality

- The AI-generated outputs were accurate and structured
- Dashboards and charts were visually clear, with consistent color coding and legends.
- Narrative summaries were concise yet insightful, making them suitable for reports and presentations.
- The outputs balanced technical details with insight summaries.

Project Enhancement

- **Faster Analysis**: Automated summarization of large datasets saved significant time compared to manual review.
- **Better Insights**: AI identified hidden patterns (e.g., weak performance in Assignments & Feedback across both courses).
- **Professional Reporting**: Generated polished dashboard visuals and structured report sections that could be directly used in academic or administrative presentations.

FIGURE-12 AI-ENHANCEDAPPROACH IN PERFORMANCE EVALUATION

Manual vs. Ai Viebuization, Dashboard, or *Content Content Produced*
basite-aesoening.form-appeoace.nnm.and

CHAPTER 5
DISCUSSION AND CONCLUSION

5.1 KEY FINDINGS

PROF. MEENA – EXCELLENCE BENCHMARK (1st-RANK)

- **RI-Excellence Score:** 15.20 (highest among peers).
- **Improvement Area Score:** 15.05, showing consistent strength across multiple teaching dimensions.
- **Rating Score:** 2.78, reflecting strong student perception and satisfaction.
- **Key Point:** Prof. Meena demonstrates balanced excellence, excelling in both core teaching delivery and improvement metrics. She sets the benchmark for faculty performance.

PROF. SURESH – RELIABLE PERFORMER (2nd- RANK)

- **RI-Excellence Score:** 15.03, above average and consistent.
- **Improvement Area Score:** 14.82, slightly lower, suggesting scope for targeted development.
- **Rating Score:** 2.93 (highest among all), highlighting strong student satisfaction and reliability.
- **Key Point:** Prof. Suresh is a dependable performer, with students rating him highly. His consistency makes him a stabilizing presence in the faculty group.

PROF. SEETHA – IMPROVEMENT-ORIENTED PERFORMER (3rd- RANK)

- **RI-Excellence Score:** 14.91, slightly lower than peers but stable.
- **Improvement Area Score:** 14.99, indicating active focus on professional growth and adaptability.
- **Rating Score:** 2.86 (highest rating score), suggesting strong student recognition of her teaching style.
- **Key Point:** While her excellence score is modest, her high rating score shows strong student engagement. She is positioned as a faculty member who is evolving and responsive to feedback.

PROF. KUMAR – IMPROVEMENT ZONE (4th- RANK)

- **RI-Excellence Score:** 14.94, lowest among peers.
- **Improvement Area Score:** 14.14, significantly lower, indicating clear areas needing support.
- **Rating Score:** 2.68, reflecting moderate student satisfaction.
- **Key Point:** Prof. Kumar requires structured mentoring and targeted interventions to improve both excellence and student perception scores. His performance is steady but below peer benchmarks

5.2 PROJECT LINKS

Raw Dataset-Student Feedback

https://docs.google.com/spreadsheets/d/1Qh5bmXE5f0NKBuZjKSnD40hVf3X2jA9D/edit?usp=sharing&ouid=103811367947603908388&rtpof=true&sd=true

Structured Feedback Dataset Link

https://docs.google.com/spreadsheets/d/11AT-DenWc2XeWZPBuqRo6DYwq-GvJRlY/edit?usp=sharing&ouid=103811367947603908388&rtpof=true&sd=true

AI –Feedback Assistant Chatbot Video link

https://drive.google.com/file/d/1rcropX4EfEfM_aZEBNo20kqpwcFqk9OL/view?usp=sharing

Github Link 0f the Project

https://github.com/drkarpagambigaik-coder/AICW-Karpagambigai.git

5.3 LIMITATIONS

The limitation of the current project is that it relies heavily on structured rating data without incorporating deeper qualitative feedback such as open-ended student comments or session-specific context, which could provide richer insights. Additionally, while AI tools like Copilot accelerated analysis and visualization, they depend on clean, well-labeled input—any ambiguity or inconsistency in the raw dataset can reduce output accuracy. The dashboard also focuses on averages and distributions, which may mask outliers or minority voices. Finally, the system is tailored to a specific course and faculty set, so broader generalization across departments or institutions would require further customization and validation.

5.4 FUTURE WORK

In the next phase, this project can be expanded by integrating qualitative feedback analysis (e.g., sentiment analysis of student comments) alongside quantitative ratings to capture richer insights. Incorporating predictive analytics in Power BI could help forecast satisfaction trends and identify at-risk themes before they decline. Another enhancement would be to connect the dashboard with real-time data sources (such as ongoing surveys or LMS platforms) so faculty and coordinators can monitor feedback continuously rather than at the end of a term. Additionally, building faculty development recommendation modules—where AI suggests targeted training or interventions based on weak areas—would make the system more actionable. Finally, scaling the framework across multiple departments or institutions, with customizable templates, would establish a standardized feedback ecosystem that supports broader educational innovation and student empowerment.

5.5 CONCLUSION

- **Prof. Meena** should be recognized as the *Excellence Benchmark*, setting standards for teaching delivery and improvement.
- **Prof. Seetha and Prof. Suresh** demonstrate strong student engagement and satisfaction, making them valuable role models for peer collaboration.
- **Prof. Kumar** requires targeted mentoring, especially in improvement areas, to align with institutional expectations.

Institutional Decision: Use the Pulse Point Ranking Board as a recurring evaluation tool to track faculty growth, reward excellence, and identify improvement zones. This ensures continuous enhancement of teaching quality and student outcomes

- **Strategic Focus:** Utilize Prof. Meena's strengths to raise institutional employability outcomes.
- **Faculty Development:** Assign Prof. Meena and Prof. Seetha as mentors in their respective strengths.
- **Strategic Focus:** Enhance classroom engagement for Prof. Kumar through targeted training.
- **Institutional Policy:** Use pulse scores to design professional development credits and recognition schemes.
- **Long-Term Vision:** Position Teaching Excellence Pulse as a recurring evaluation tool to sustain high standards and foster collaborative growth.

- **Balanced Benchmark:** Position Prof. Suresh as a role model for steady performance.
- **Development Path:** Provide Prof. Kumar with structured training and peer mentoring
- **Policy Integration:** Make Learning Outcome Pulse a recurring evaluation tool to ensure continuous improvement in both growth and employability.

TABLE-8 PULSE POINT-FACULTY PERFORMANCE RANKING BOARD

Faculty	RI-Excellence Score	Improvement Area Score	Rating Score	Ranking Score	Final Rank	Key Findings
Prof. Meena (B.Com-Gen)	15.20	15.05	2.78	**33.03**	1st-Rank	Excellence benchmark – consistently high across all metrics. Strong syllabus mastery, balanced improvement, and positive student ratings.
Prof. Suresh (B.Com-CA)	15.03	14.82	2.93	**32.76**	2nd-Rank	Reliable performer – highest student rating score, showing strong student satisfaction. Balanced but slightly lower improvement score.
Prof. Seetha (B.Com-AF)	14.91	14.99	2.86	**32.78**	3rd-Rank	Improvement-oriented – strong adaptability and student recognition. Slightly lower excellence score but high student engagement.
Prof. Kumar (B.Com-CS)	14.94	14.14	2.68	**31.76**	4th-Rank	Improvement zone – lowest improvement area score and rating score. Needs targeted mentoring and support to align with peer benchmarks.

.This project successfully transformed raw student feedback data into actionable insights through a structured pipeline of data ingestion, cleaning, analysis, visualization, and AI integration. By combining Excel formulas, Power BI dashboards, and Copilot's AI-driven summaries, the system

provided a clear picture of faculty performance, course satisfaction, and theme-specific strengths and weaknesses.

AI integration significantly reduced analysis time, improved reporting quality, and delivered presentation-ready outputs. Although the current approach is limited by its reliance on structured ratings and course-specific scope, it establishes a strong foundation for future expansion into qualitative feedback analysis, predictive modeling, and institution-wide deployment. Ultimately, the project demonstrates how data-driven insights and AI support can empower educators to make informed decisions, enhance teaching effectiveness, and improve student learning experiences.

REFERENCES

[1] Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel

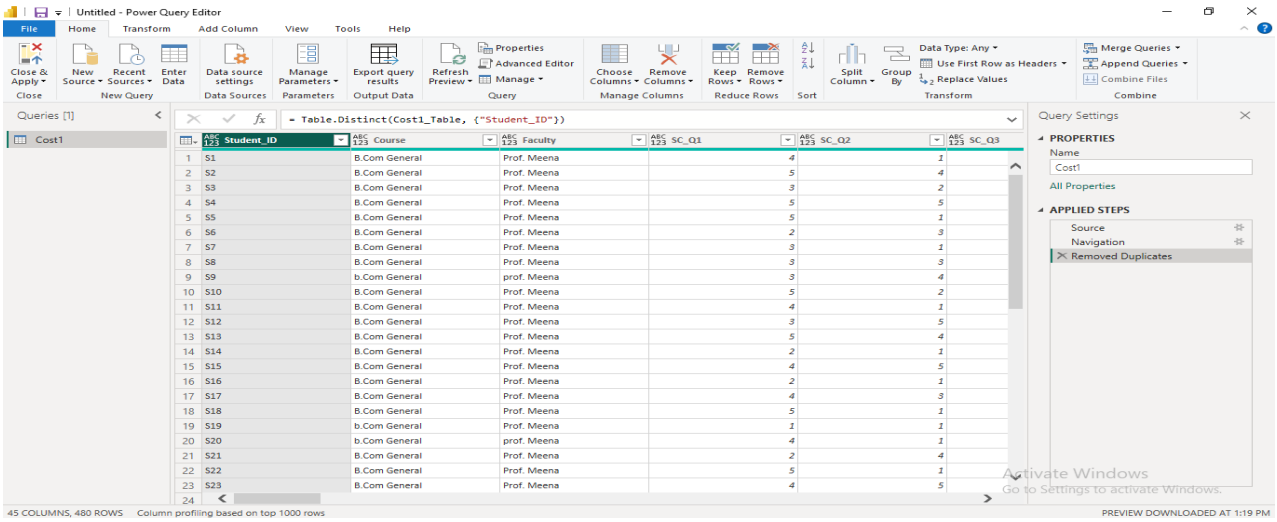
[2] Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/

[3] Microsoft Corporation. Microsoft Copilot — AI-Powered Productivity. Available at: https://copilot.microsoft.com

[4] Edunet Foundation. AICW Fellowship Program Guidelines. 2026.

APPENDICES

Appendix A — Sample Dataset



Student_ID	Course	Faculty	SC_Q1	SC_Q2	SC_Q3
S1	B.Com General	Prof. Meena	4	1	
S2	B.Com General	Prof. Meena	5	4	
S3	B.Com General	Prof. Meena	3	2	
S4	B.Com General	Prof. Meena	5	5	
S5	B.Com General	Prof. Meena	5	1	
S6	B.Com General	Prof. Meena	2	3	
S7	B.Com General	Prof. Meena	3	1	
S8	B.Com General	Prof. Meena	3	3	
S9	b.Com General	prof. Meena	3	4	
S10	B.Com General	Prof. Meena	5	2	
S11	B.Com General	Prof. Meena	4	1	
S12	B.Com General	Prof. Meena	3	5	
S13	B.Com General	Prof. Meena	5	4	
S14	B.Com General	Prof. Meena	2	1	
S15	B.Com General	Prof. Meena	4	5	
S16	B.Com General	Prof. Meena	2	1	
S17	B.Com General	Prof. Meena	4	3	
S18	B.Com General	Prof. Meena	5	1	
S19	b.Com General	Prof. Meena	1	1	
S20	b.Com General	prof. Meena	4	1	
S21	B.Com General	Prof. Meena	2	4	
S22	B.Com General	Prof. Meena	5	1	
S23	B.Com General	Prof. Meena	4	5	

Appendix B — Key Excel Formulas / Measures Used

Teaching Quality

Teaching_Quality = DIVIDE(’Table’[SC_Avg] + ’Table’[CM_Avg] + ’Table’[AF_Avg], 3)

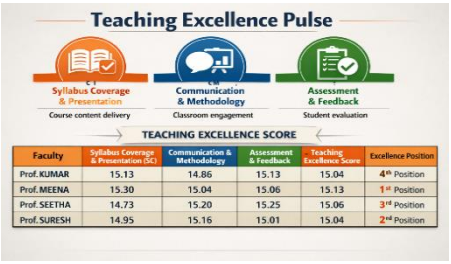
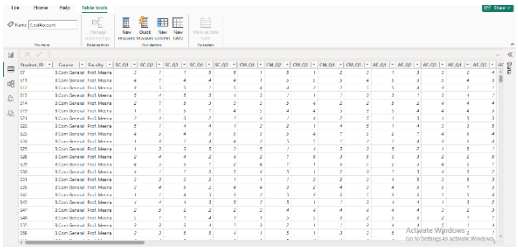
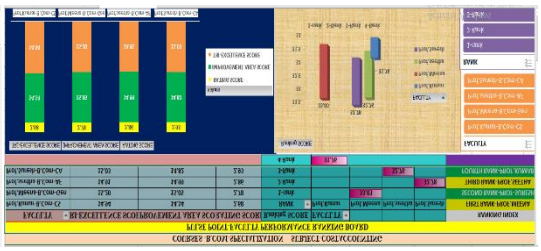
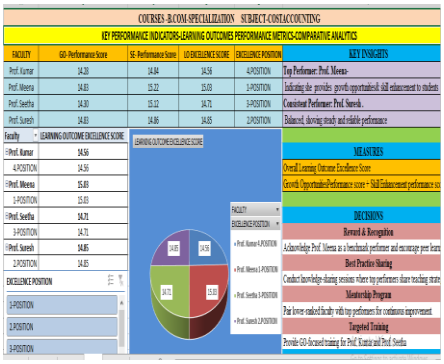
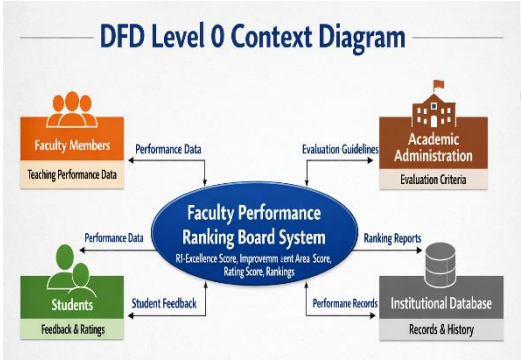
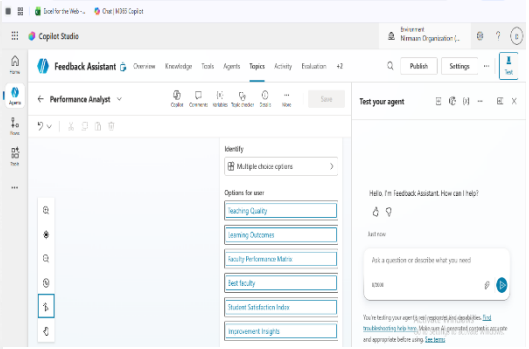
Learning Outcome

Learning_Outcome = DIVIDE(’Table’[GO_Avg] + ’Table’[SE_Avg], 2)

Student Experience

Student_Experience = DIVIDE(’Table’[TE_Avg] + ’Table’[SF_Avg], 2)

Appendix C — AI Prompts Used

- **A. Comparative Analysis Prompts-**“Compare Teaching Excellence Scores across faculties - highlight performance gaps.”
- **B. Diagnostic Insight Prompts-** “Analyze the Scores of the Learning outcomes of the students based on Faculty Performance Metrics”
- **C. Predictive / Prescriptive Prompts-**“Suggest improvement strategies for faculties with lower satisfaction scores.”
- **Descriptive Analysis Prompts-**“Summarize the overall student satisfaction levels across all faculties.”

Appendix D — Screenshot Gallery